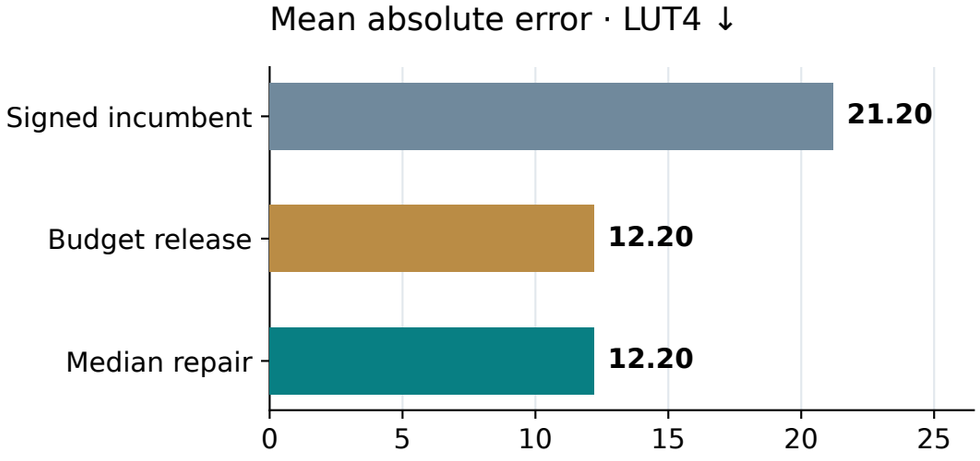
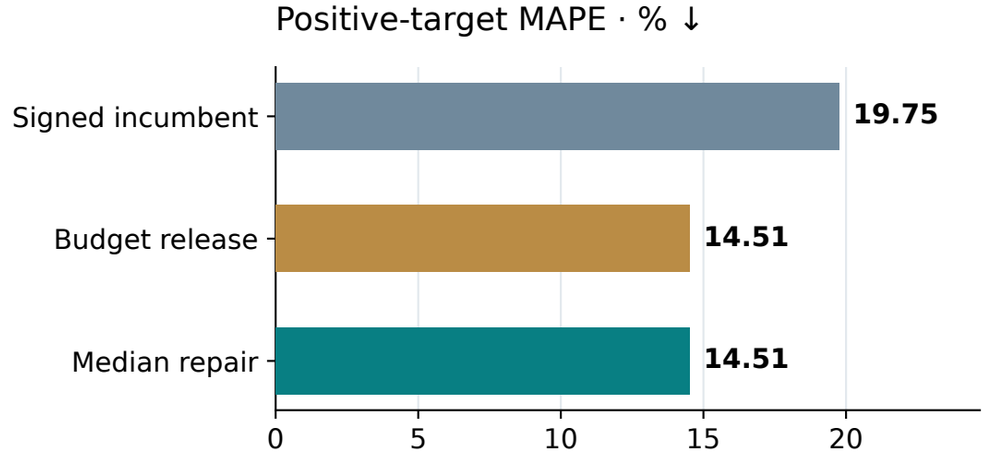
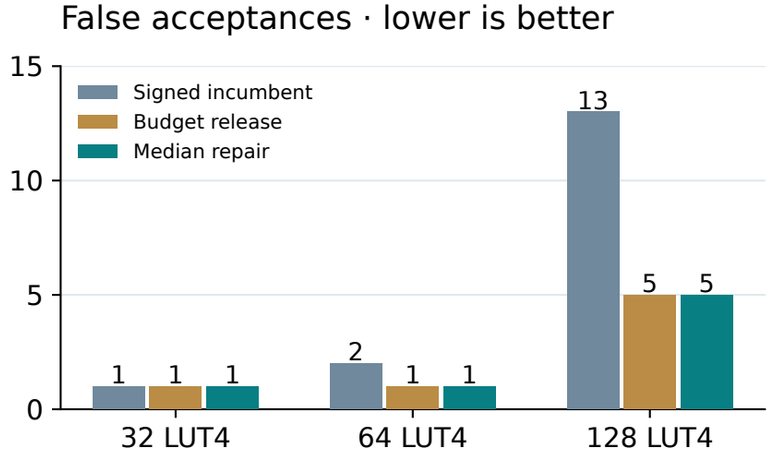
Repairing a budget-boundary error without retraining

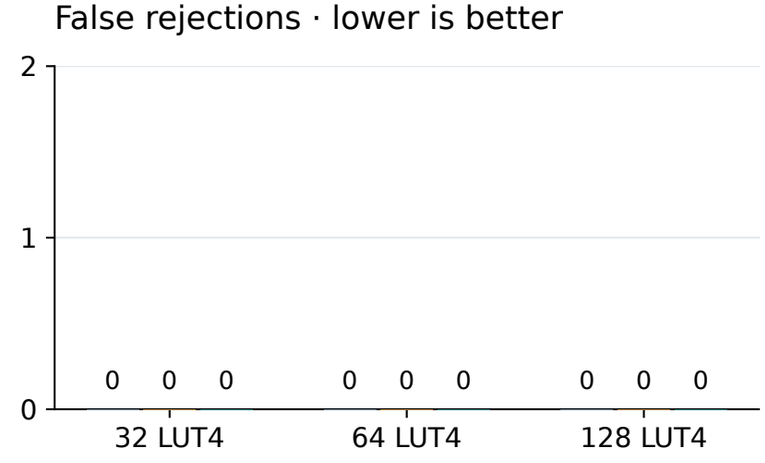
40 fresh synthetic circuits · matched labels · policies and predictions frozen before test synthesis

Predeclared selection and fresh gates: NOT PASSED — repair changed 0 fresh predictions









Consumed boundary case: actual 125 LUT4; release 129.32 → repair 128.00 at cap128.

32 LUT4: 2 feasible / 38 infeasible · 64 LUT4: 13 feasible / 27 infeasible · 128 LUT4: 21 feasible / 19 infeasible

Worst underprediction (signed → release → repair): 93.00 → 41.22 → 41.22 LUT4.

One frozen median rule; zero new fits. Prior evaluation panels are consumed development evidence, separate from this fresh cohort.

Targeted isolated iCE40 area study; no calibrated safety, application-transfer, timing, power or CPU-speed claim.